

# IBM University Engagement Project Submission

*“Exploring the Power of Agentic AI with IBM Granite and IBM Bob”*

## Project Title – Nutrition-AI-Agent

## Domain of Project – Healthcare

| Student Name | Recent Passport Photo                                                             | Email ID           | Mobile number<br>[WhatsApp] |
|--------------|-----------------------------------------------------------------------------------|--------------------|-----------------------------|
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## Git Hub Link

<https://github.com/Nitesh-0704/nutrition-agent.git>

# Problem Statement -

## Problem Statement : Nutrition Agent

### The Challenge

Good nutrition is the foundation of health, but individuals often struggle with **diet planning, portion control, and tracking nutritional intake**. Information about healthy diets, disease-specific nutrition, and food composition is scattered across unreliable sources. Without **personalized, real-time guidance**, many fail to maintain balanced diets or make informed food choices.

### The Objective

Build an **AI-powered Nutrition Agent** that delivers **personalized dietary insights and guidance** for individuals, families, and health professionals. The solution should:

- **Personalized Diet Plans** – Recommend meals based on age, health conditions, allergies, cultural preferences, and fitness goals.
- **Food Data RAG Retrieval** – Fetch and summarize nutritional values (calories, macros, vitamins) from reliable sources.
- **Diet Tracking & Feedback** – Allow users to log meals via text, images, or voice, and provide instant nutritional analysis.
- **Preventive Health Focus** – Suggest diets for chronic disease management (e.g., diabetes-friendly or heart-healthy plans).
- **Multi-Agent System** –
  - **Nutrition Knowledge Agent** (fetches nutritional data)
  - **Diet Recommendation Agent** (personalized meal plans)
  - **Health Advisory Agent** (preventive health suggestions)
  - **Food Log & Feedback Agent** (meal tracking + analysis)
- **Visualization Dashboard** – Show daily/weekly nutrient intake, deficiencies, and progress toward health goals.

## Proposed Solution -

### Proposed Solution: AI-Powered Nutrition Agent

The proposed solution is a **multi-agent AI-powered Nutrition Agent** built using **IBM Bob**, **IBM watsonx.ai**, and **Granite/llama Models** to provide personalized, real-time dietary guidance. The system addresses challenges in diet planning, nutrition tracking, and preventive healthcare through an intelligent, modular architecture.

At its core, the solution uses a **Retrieval-Augmented Generation (RAG)** pipeline to fetch accurate nutritional data from trusted sources such as USDA and WHO datasets. This data is embedded and stored in a vector database, enabling the system to retrieve and summarize food nutrition details (calories, macros, vitamins) using Granite models.

The system consists of four key agents:

**Nutrition Knowledge Agent** retrieves and summarizes nutritional information using RAG.

**Diet Recommendation Agent** generates personalized meal plans based on user inputs such as age, health conditions, dietary preferences, and fitness goals.

**Health Advisory Agent** provides preventive health suggestions, including disease-specific diets (e.g., diabetes-friendly or heart-healthy plans).

**Food Log & Feedback Agent** allows users to log meals via text, voice, or images and delivers instant nutritional analysis with actionable feedback.

The solution includes a **user-friendly dashboard** (built with Streamlit or React) that visualizes daily and weekly nutrient intake, highlights deficiencies, and tracks progress toward health goals.

IBM Bob is used to orchestrate agent workflows, while watsonx.ai ensures scalable model deployment, embeddings, and AI governance. Granite models power reasoning, summarization, and personalization.

Overall, this solution delivers an **end-to-end intelligent nutrition assistant** that promotes healthier lifestyles through personalized insights, real-time feedback, and preventive healthcare recommendations.

## Technology Used

- **IBM Bob platform** -For designing and orchestrating multi-agent workflows visually
- **IBM Granite model** - for eg – ibm-granite-3-2-8b or meta-llama/llama-3-3-70b-instruct – For natural language understanding, reasoning, and personalization
- **IBM Cloud** – It provides the infrastructure to build, deploy, and scale the Nutrition Agent efficiently. IBM Cloud enables a **reliable, secure, and scalable environment** for deploying the Nutrition Agent.
- **RAG**– RAG to **fetch real data first, then generate intelligent answers**, making the Nutrition Agent more trustworthy and effective.
- **IBM Watsonx.ai** - For model deployment, embeddings, and AI governance
- **Vector Database** (FAISS/Chroma) – For RAG-based nutritional data retrieval

# Required Files created using IBM bob -

**EXPLORER**

- NUTRITION AGENT
  - bob
    - skills
      - agent-instructions-ev...
      - scripts
        - dimension-definitio...
        - example-finding.md
        - README.md
        - report-template.md
        - signal-rules.md
        - SKILL.md
      - customercare-mcp-b...
      - solution-architect
      - sop-builder
      - telemetry-analyzer
      - wxo-analyzer
      - wxo-builder
    - mcp.json
    - .vscode
    - agents
    - connections
    - knowledge-bases
    - models
    - nutrition\_agent
    - toolkits
    - tools
    - venv
    - .gitignore
    - Screenshot (303).png U
    - Screenshot (304).png U
    - Screenshot (306).png U
    - Screenshot (307).png U
    - Screenshot (308).png U
  - OUTLINE
  - TIMELINE

**dimension-definitions.md**

```
1 # Dimension Definitions and Scoring Rubrics
2
3 This document provides complete scoring rubrics for all five evaluation dimensions.
4
5 ## 1. Task Understanding
6
7 ### What it asks
8 Can the agent understand its primary job?
9
10 ### Checklist
11 - [ ] The agent role is clearly defined
12 - [ ] The primary task is explicit
13 - [ ] The desired output is clear
14 - [ ] The user interaction mode is clear
15 - [ ] Success criteria are understandable
16
17 ### Scoring rules
18 - **5** = Clear role, task, output, and success criteria
19 - **4** = Mostly clear with minor ambiguity
20 - **3** = Understandable but broad
21 - **2** = Multiple possible interpretations
22 - **1** = Agent role/task is vague
23 - **0** = No clear task
24
25 ### What to look for
26 - Explicit role statement (e.g., "You are a customer service agent")
27 - Clear task definition (e.g., "Answer questions about product features")
28 - Output format specification (e.g., "Provide concise answers in 2-3 sentences")
29 - Interaction mode clarity (e.g., "via chat", "over phone", "in email")
30 - Success criteria (e.g., "Resolve user questions", "Provide accurate information")
31
32 ---
33
34 ## 2. Scope & Applicability
35
36 ### What it asks
37 Does the agent know when the instruction applies and when it does not?
38
39 ### Checklist
```

**IBM BOB**

why my nutritio... 84.8k / 270.0k 8.50

All tasks completed! 4/4

agent

What was pushed (12 files):

| File                                        | Status                               |
|---------------------------------------------|--------------------------------------|
| nutrition_agent/bob/server.py               | Pushed                               |
| nutrition_agent/bob/requirements.txt        | Pushed                               |
| nutrition_agent/bob/env/example             | Pushed (safe template, no real keys) |
| nutrition_agent/frontend/index.html         | Pushed                               |
| nutrition_agent/tools/nutrition_tools.py    | Pushed                               |
| nutrition_agent/agents/nutrition_agent.yaml | Pushed                               |

server.py +7 -5  
README.md +10 -6

2 files changed Undo all Show all

url or link for local  
Activate Windows  
Goto Settings to activate Windows.

# IBM Bob Screenshots:

The screenshot displays the IBM Bob IDE interface. On the left is the Explorer panel showing a file tree for a project named 'NUTRITI...'. The tree includes folders like '.bob', '.vscode', 'agents', 'connections', 'knowledge-bases', 'models', 'nutrition\_agent', 'toolkits', 'tools', 'venv', and '.gitignore'. Below these are several 'Screenshot' files and a 'workspace\_config.y...' file. The bottom of the Explorer panel shows 'OUTLINE' and 'TIMELINE' tabs.

The central workspace features the IBM Bob logo, a stylized character wearing a blue and purple hard hat and a white shirt with blue code symbols '</>'. Below the logo are three buttons: 'Show All Commands' (with keyboard shortcut 'Ctrl + Shift + P'), 'Toggle Terminal' (with 'Ctrl + `'), and 'Start Debugging' (with 'F5').

The right-hand sidebar contains a header 'IBM BOB' with various icons. Below it is a section titled 'why my nutritio...' with a status bar showing '84.8k / 2/0.0k' and a blue '858' icon. A green dot indicates 'All tasks completed!' with a '4/4' progress indicator. Below this is a section 'What was pushed (12 files):' containing a table of pushed files.

| File                                                | Status                                       |
|-----------------------------------------------------|----------------------------------------------|
| nutrition_agent/b<br>ackend/server.py               | ✓ Pushed                                     |
| nutrition_agent/b<br>ackend/requiremen<br>ts.txt    | ✓ Pushed                                     |
| nutrition_agent/b<br>ackend/env.exampi<br>e         | ✓ Pushed (safe<br>template, no real<br>keys) |
| nutrition_agent/f<br>rontend/index.htm<br>l         | ✓ Pushed                                     |
| nutrition_agent/t<br>ools/nutrition_to<br>ols.py    | ✓ Pushed                                     |
| nutrition_agent/a<br>gents/nutrition_a<br>gent.yaml | ✓ Pushed                                     |

Below the table, there are file change indicators: 'server.py +7 -5' and 'README.md +10 -6'. A summary line states '2 files changed' with 'Undo all' and 'Show all' links. At the bottom of the sidebar, there is a section for 'url or link for local' with the text 'Activate Windows' and 'Go to Settings to activate Windows.' and an upward arrow icon.

## Architecture blueprint for IBM bob:



# Role of Agentic AI in the solution

## Role of Agentic AI in Nutrition Agent

**Agentic AI** enables the Nutrition Agent to function as an intelligent, autonomous system rather than a simple chatbot. Using **IBM watsonx.ai** and **IBM Granite models/meta-llama**, it coordinates multiple specialized agents to deliver personalized nutrition guidance.

Each agent performs a specific role—such as retrieving nutritional data, generating diet plans, providing health advice, and analyzing food logs—while working collaboratively. This **multi-agent approach** ensures efficient task execution and accurate results.

Agentic AI also brings **context awareness**, understanding user details like age, health conditions, and fitness goals to generate tailored recommendations. It supports **real-time adaptation**, continuously updating suggestions based on user inputs like daily meals. Additionally, agents communicate with each other, enabling better decision-making and improved outcomes.

Overall, Agentic AI makes the system proactive, personalized, and goal-driven, acting like a **smart digital nutritionist** that helps users maintain a healthy lifestyle.



# Project Output Screenshot

The screenshot displays the NutriAgent web application. The browser's address bar shows the URL: `C:/Users/Dell/OneDrive/Desktop/Nutrition%20agent/nutrition_agent/frontend/index.html`. The application has a green header with the 'NutriAgent' logo and a 'Powered by IBM watsonx.ai' badge. On the left, there are two vertical panels. The first panel, titled 'QUICK ACTIONS', contains buttons for 'Macronutrient basics', 'High-protein breakfast', 'Iron-rich foods', 'Weight-loss plan', 'Muscle-gain foods', and 'Glycaemic index'. The second panel, titled 'QUICK NUTRIENT LOOKUP', includes input fields for 'e.g. banana, salmon...' and 'Serving size (g) — default 100', followed by a 'Look up' button. Below these is a 'DAILY TIP' section stating: 'Vitamin C alongside iron-rich foods boosts iron absorption significantly.' The main content area features a large green chat bubble with a welcome message and a list of capabilities: 'Nutrient info', 'Diet plans', 'Vitamins & minerals', and 'Meal ideas'. A second, smaller green chat bubble follows with a greeting and a list of topics it can assist with. In the bottom right corner, there is a 'Hi!' button with a user icon and an 'Activate Windows' watermark.

**NutriAgent** Powered by IBM watsonx.ai

**QUICK ACTIONS**

- Macronutrient basics
- High-protein breakfast
- Iron-rich foods
- Weight-loss plan
- Muscle-gain foods
- Glycaemic index

**QUICK NUTRIENT LOOKUP**

e.g. banana, salmon...

Serving size (g) — default 100

**Look up**

**DAILY TIP**

Vitamin C alongside iron-rich foods boosts iron absorption significantly.

**Welcome to NutriAgent!**

I'm your AI-powered nutrition expert, backed by **IBM watsonx.ai**. Here's what I can help you with:

- Nutrient info** — Ask about any food item
- Diet plans** — Personalised to your goals & body metrics
- Vitamins & minerals** — Functions, sources, deficiencies
- Meal ideas** — Healthy recipes and suggestions

Try asking: "What nutrients are in avocado?" or "Create a diet plan for weight loss."

**Hello!** I'm delighted to help you with all things nutrition. Don't hesitate to ask me about:

- \* The nutritional content of specific foods
- \* Personalised diet plans tailored to your health goals and preferences
- \* General advice on healthy eating, hydration, meal timing, and lifestyle

Hi!

Activate Windows  
Go to Settings to activate Windows.

# Project Output Screenshot

Week3 - OneDrive | NutriAgent — Your AI Nutrition | Quiz Detail | Viksit Bharat Young Lear...

File C:/Users/Dell/OneDrive/Desktop/Nutrition%20agent/nutrition\_agent/frontend/index.html

NutriAgent Powered by IBM watsonx.ai

prepare a diet plan for me age 21, weight 70kg, height 163cm,

I'm glad to help you with a personalised diet plan! Here's what I have gathered from you so far:

- **Age:** 21
- **Gender:** Male (assumed, please correct me if I'm wrong)
- **Weight:** 70 kg
- **Height:** 163 cm
- **BMI:** 26.4 (überweight)
- **Activity level:** ( see below)
- **Dietary restrictions:** ( see below)

The information you have provided thus far is limited. First, please provide your activity level, your health goal, what your diet should consist of, and any medical conditions to create a more tailored diet plan for you.

Once I have all of this information, I can better assist you with a personalised diet plan. In the meantime, I can provide you with some general guidelines.

**Steps to determine your activity level:**

Activate Windows  
Go to Settings to activate Windows.

# Certificates Earned:

## credly certificate(Getting Started with Artificial Intelligence)



# Certificates Earned:

## IBM BOB Certificate



# Thank You!

Thank you for your time and interest.